

ARDUINO & HC-SR04 ULTRASONIC SENSOR

Comprehensive Lesson Guide & Practical Implementation

Real-World Scenario: How Do Cars & Auto-Rickshaws Know When to Stop?

Have you ever seen a car stop just before hitting a wall while parking? Imagine Rahul's uncle parking his auto-rickshaw in a tight Bengaluru lane. How does he avoid bumping into obstacles? Vehicles use proximity sensing technology powered by sound waves!

Fun Fact: Bats and dolphins use echolocation to 'see' and navigate in the dark using sound waves exactly like the HC-SR04 sensor!

1. Introduction & Required Hardware Materials

To build our working distance measurement system, we require the following core hardware components:

Hardware Materials Needed:

- Arduino Uno Board:** The main microcontroller that acts as the brain, running our logic code and measuring timing.
- HC-SR04 Ultrasonic Distance Sensor:** Measures distances from 2 cm to 400 cm with high accuracy using 40 kHz sound waves.
- Jumper Wires (Male-to-Female / Male-to-Male):** Establishes electrical connections between the Arduino pins and sensor terminals.
- Optional Alert Output Components:** Red LED, Buzzer, 220Ω resistor, and a Breadboard.

HC-SR04 Sensor Pinout & Arduino Connection Table

Pin Name	Arduino Connection	Function Description
VCC	5V Pin	Provides +5V power supply to the ultrasonic sensor module.
GND	GND Pin	Ground connection to complete the electrical circuit.
Trig (Trigger)	Digital Pin 2	Input pin on sensor; receives a 10μs pulse from Arduino to emit sound waves.
Echo	Digital Pin 3	Output pin on sensor; sends high signal to Arduino while waiting for reflected sound.

2. Physics of Measurement: How It Works

The sensor transmits a short 40 kHz ultrasonic pulse through air. When the wave encounters an obstacle, it bounces back (echoes) to the receiver.

$$\text{Distance} = (\text{Speed of Sound} \times \text{Time}) \div 2$$

Why divide by 2? Sound travels from the transmitter TO the object and then reflects BACK to the receiver. That is two full trips for one measurement! Dividing by two isolates the one-way distance.

Key Constant: Speed of Sound in Air ≈ 343 m/s (or **0.034 cm/ μ s**).

3. Hands-On Hardware & Coding Step-by-Step Guide

Step 1: Circuit Assembly & Wiring Setup

Using your **jumper wires**, connect the **HC-SR04 sensor** to the **Arduino** board as follows:

- Connect VCC on HC-SR04 to Arduino **5V** pin using a jumper wire.
- Connect GND on HC-SR04 to Arduino **GND** pin using a jumper wire.
- Connect Trig on HC-SR04 to Arduino **Digital Pin 2** using a jumper wire.
- Connect Echo on HC-SR04 to Arduino **Digital Pin 3** using a jumper wire.

Step 2: Configure Arduino Pins in Code

Inside the `setup()` function, configure Pin 2 as OUTPUT (Trig) and Pin 3 as INPUT (Echo). Initialize serial communication at 9600 baud rate.

Step 3: Send an Ultrasonic Sound Pulse

Set Trig LOW for 2μs, then HIGH for 10μs to emit an ultrasonic burst, and return it to LOW.

Step 4: Read Time & Calculate Distance in cm

Use `pulseIn(echoPin, HIGH)` to measure duration in microseconds. Calculate distance using formula:
 $\text{Distance (cm)} = (\text{duration} * 0.034) / 2.$

Step 5: Output to Serial Monitor & Logic Control

Display real-time measurements on screen. Add logic control (e.g., if distance < 15 cm, trigger alert system).

```
// Complete Arduino Code
const int trigPin = 2;
const int echoPin = 3;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  float distance = (duration * 0.034) / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
  delay(100);
}
```

STUDENT ASSESSMENT & PRACTICE WORKSHEET

Test your understanding of the HC-SR04 Ultrasonic Sensor & Arduino Circuit

REQUIRED LAB MATERIALS CHECKLIST:

- **Arduino Board** (Microcontroller) • **HC-SR04 Ultrasonic Sensor** (Transmitter & Receiver)
- **Jumper Wires** (4x Circuit Interconnects) • **USB Cable & Computer** (For power & programming)

Question 1 (Multiple Choice)

Which pin on the HC-SR04 sensor is responsible for sending out the ultrasonic sound wave pulse?

- ☐ A) VCC
- ☐ B) Trig
- ☐ C) Echo
- ☐ D) GND

Question 2 (Multiple Choice)

Why is the total measured travel time of the ultrasonic wave divided by 2 in the distance calculation formula?

- ☐ A) To convert microseconds into milliseconds
- ☐ B) Because sound travels at twice the normal speed in air
- ☐ C) Because the wave travels to the obstacle and back, covering double the distance
- ☐ D) To account for the voltage drop across the sensor pins

Question 3 (Multiple Choice)

Ultrasonic sound waves operate in a frequency range that is:

- ☐ A) Directly audible to human ears
- ☐ B) Below 20 Hz (Infrasonic)
- ☐ C) Above 20 kHz (Inaudible to humans)
- ☐ D) Identical to FM radio frequency waves

Question 4 (Multiple Choice)

If the `pulseIn()` function records a travel time of 1000 microseconds, what is the calculated distance to the object using $\text{speed} = 0.034 \text{ cm}/\mu\text{s}$?

- ☐ A) 34 cm
- ☐ B) 17 cm
- ☐ C) 68 cm
- ☐ D) 8.5 cm

Question 5 (Multiple Choice)

In an automated obstruction detection code, if an object is detected at 10 cm and the threshold condition is set to `if (distance < 15)`, what action will the system execute?

- ☐ A) Keep the alert LED and buzzer deactivated
- ☐ B) Activate the alert system (turn ON LED and Buzzer)
- ☐ C) Reset the Arduino controller
- ☐ D) Disconnect power to the VCC pin

Question 6 (Multiple Choice)

What mode must the Arduino pin connected to the sensor's Echo pin be set to using the pinMode() function?

- ☐ A) OUTPUT
- ☐ B) INPUT
- ☐ C) HIGH
- ☐ D) LOW

Question 7 (Hands-On Hardware Identification)

Match each required lab material to its corresponding function in your circuit setup:

Material	Function in Circuit
1. Arduino Board	Executes timing calculations and processes trigger/echo logic signals
2. HC-SR04 Sensor	Emits 40kHz ultrasonic waves and detects returning echo pulses
3. Jumper Wires	Carries 5V, GND, Trig, and Echo electrical signals between board and sensor

Mini Challenge: Design Task

Scenario: You are tasked with designing a Monsoon Flood Early Warning System using the Arduino, HC-SR04 ultrasonic sensor, and jumper wires [cite: 4].

Your Task: Write out the circuit wiring connections using your jumper wires and explain the logical flow (pseudocode or block plan) for positioning the sensor above a drainage canal to trigger emergency alerts when rising water levels pass critical height thresholds [cite: 4].